

Seshadri Constants and Infinitesimal Polygons: The Kollár–Lazarsfeld Abelian Surface Example

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1 Setup and claim structure, keeping close to the handwritten draft

1.1 Basic setup from the handwritten draft

Let E be an elliptic curve over $\mathbb{C}$, and set

$$S := E \times E.$$

Let $F_1, F_2 \subset S$ be fibres of the two projections and let $\Delta \subset S$ be the diagonal. Fix an integer $n \geq 2$ and use the Kollár–Lazarsfeld divisor

$$A_n := nF_1 + (n^2 - n + 1)F_2 - (n - 1)\Delta.$$

Thus A_n is an integral Cartier divisor. With the standard intersection numbers on $E \times E$,

$$F_1^2 = F_2^2 = \Delta^2 = 0, \quad F_1 \cdot F_2 = F_1 \cdot \Delta = F_2 \cdot \Delta = 1,$$

one has

$$A_n^2 = 2, \quad A_n \cdot (F_1 + F_2) = n^2 - 2n + 3 > 0.$$

Therefore A_n is ample, by Lazarsfeld’s Lemma/Example 1.5.4 as used in Example 1.5.7. Set

$$R := F_1 + F_2.$$

Let

$$B \in |2R| = |2(F_1 + F_2)|$$

be a smooth divisor, and let

$$\pi: X \rightarrow S = E \times E$$

be the standard double cover associated to the line bundle $\mathcal{O}_S(R)$ and the branch section cutting out B . Thus

$$\pi_*\mathcal{O}_X \simeq \mathcal{O}_S \oplus \mathcal{O}_S(-R).$$

Since S and B are smooth and the characteristic is zero, the standard local form $w^2 = v$ shows that X is a smooth projective surface. Finally set

$$D_n := \pi^*A_n.$$

Then D_n is ample because π is finite and A_n is ample.

Choose a point

$$x \in S \setminus B.$$

Then

$$\pi^{-1}(x) = \{y_1, y_2\}.$$

Let

$$\pi_x: \text{Bl}_x(E \times E) \rightarrow E \times E$$

be the blow-up at x , with exceptional divisor E_x . Let

$$\pi_1: \text{Bl}_{y_1}(X) \rightarrow X$$

be the blow-up at y_1 , with exceptional divisor E_1 , and let

$$\pi_{1,2}: \text{Bl}_{y_1, y_2}(X) \rightarrow X$$

be the blow-up at both y_1 and y_2 . Since $x \notin B$, the map π is étale at y_1 and y_2 .

Throughout the global rigorous discussion below, we work over $\mathbb{C}$, and all points under consideration are closed points. The characteristic $\neq 2$ hypothesis reappears only in the formal local branch calculation. In the present Kollár–Lazarsfeld setup, characteristic zero and the smoothness of B already guarantee smoothness of the double cover X .

$$\begin{array}{ccccc}
\pi_x^* A_n & E_x & \longleftarrow & \tilde{E}_{y_1}, \tilde{E}_{y_2} & \tilde{\pi}^* \pi_x^* A_n \\
& & & \cap & \parallel \\
\text{Bl}_x(E \times E) & \xleftarrow[\text{2-1}]{\tilde{\pi}} & & \text{Bl}_{y_1, y_2}(X) & \pi_{1,2}^* D_n \\
\downarrow & & & \downarrow \pi_2 & \\
& & & E_1 \subseteq \text{Bl}_{y_1}(X) & \pi_1^* D_n \\
& & & \downarrow \pi_1 & \\
A_n & & & & \pi^* A_n = D_n \\
\downarrow & & & & \\
x \in E \times E & \xleftarrow[\text{2-1}]{\pi} & & X & \ni y_1, y_2 = \pi^{-1}(x) \\
\cup & & & & \\
x \notin B & \in & & |2(F_1 + F_2)| &
\end{array}$$

1.2 The two main claims that actually drive the handwritten draft

The handwritten page contains many calculations and reminders, but only two genuine target claims stand out. The first one is written explicitly in the red box at the upper right of the page; the second one is the lower positivity goal suggested by the polygon sketch and the notation ε'_x and ε_x . The other statements are better viewed as *supporting inputs* for these two claims, rather than as separate main claims.

Claim 1.1 (the second blow-up does not change the infinitesimal polygon over y_1). *Choose a point $z_1 \in E_1$ and let $\tilde{z}_1 \in \tilde{E}_{y_1}$ be the corresponding point under the identification induced by the second blow-up isomorphism near E_1 . Then there is a canonical equality*

$$\Delta_{(\tilde{E}_{y_1}, \tilde{z}_1)}(\pi_{1,2}^* D_n) = \Delta_{(E_1, z_1)}(\pi_1^* D_n).$$

Role in the draft. This is the boxed claim written on the manuscript. It isolates the fact that the second blow-up at y_2 should not affect the infinitesimal polygon attached to the flag over y_1 .

Claim 1.2 (largest inverted simplex constant and Seshadri constant). *Fix $z \in E_x$ and define the largest inverted simplex constant in the notation of the manuscript*

$$\varepsilon'_x := \sup \left\{ \xi \geq 0 \mid \Delta_\xi^{-1} \subseteq \Delta_{(E_x, z)}(\pi_x^* A_n) \right\}.$$

Here

$$\Delta_\xi^{-1} := \{(t, y) \in \mathbb{R}_{\geq 0}^2 \mid 0 \leq t \leq \xi, 0 \leq y \leq t\}$$

is the two-dimensional inverted simplex of size ξ . Also define

$$\mu_x := \sup \{t \mid \pi_x^* A_n - tE_x \text{ is big}\}, \quad \varepsilon_x := \sup \{t \mid \pi_x^* A_n - tE_x \text{ is ample}\}.$$

Then the main target suggested by the draft is

$$\varepsilon'_x = \varepsilon_x.$$

Equivalently, since A_n is ample, this is the same as proving

$$\varepsilon'_x = \varepsilon(A_n; x).$$

Role in the draft. The handwritten page already indicates the easy direction $\varepsilon'_x \leq \varepsilon_x$, and the real issue is how to obtain the opposite direction. The rigorous proof given later reduces this to the standard identification between infinitesimal polygons and Seshadri constants on surfaces.

1.3 Supporting inputs extracted from the draft

The following auxiliary statements appear naturally in the handwritten strategy and will be used later without being treated as separate main claims.

Claim 1.3 (étale-local comparison for the pullback graded linear series). *Let*

$$W_m := \pi^* H^0(E \times E, \mathcal{O}_{E \times E}(mA_n)) \subseteq H^0(X, \mathcal{O}_X(mD_n)), \quad W_\bullet := \bigoplus_{m \geq 0} W_m.$$

Here $W_\bullet$ is the pullback graded linear series associated to A_n and π ; this notation is local to this note.

Then, after choosing corresponding infinitesimal flags over x and y_1 , one has

$$\nu_{(E_1, z_1)}(\pi^* s) = \nu_{(E_x, z)}(s) \quad \text{for every } 0 \neq s \in H^0(E \times E, \mathcal{O}_{E \times E}(mA_n)),$$

where $z \in E_x$ and $z_1 \in E_1$ correspond under the isomorphism $\mathbb{P}(T_{y_1} X) \simeq \mathbb{P}(T_x S)$ induced by $d\pi_{y_1}$. Consequently,

$$\Delta_{(E_x, z)}(\pi_x^* A_n) = \Delta_{(E_1, z_1)}(W_\bullet) \subseteq \Delta_{(E_1, z_1)}(\pi_1^* D_n).$$

How it is used. This is the rigorous form of the manuscript’s informal sentence “calculate the polygon on X from the polygon on S ”. What is justified is equality for the pullback graded linear series $W_\bullet$, hence an inclusion into the Newton–Okounkov polygon of the complete linear series on X .

Claim 1.4 (the negative part vanishes below the Seshadri constant). *Let*

$$L_t := \pi_x^* A_n - tE_x, \quad L_t = P_t + N_t$$

be the Zariski decomposition on $\mathrm{Bl}_x(E \times E)$. If

$$0 \leq t < \varepsilon_x,$$

then

$$N_t = 0.$$

How it is used. This auxiliary step formalizes the lower part of the handwritten note: below the Seshadri constant there is no negative part, so the sketch of the infinitesimal polygon is controlled by the nef divisor itself.

Claim 1.5 (the exceptional divisor is not a component of the negative part). *Let*

$$L_t := \pi_x^* A_n - tE_x, \quad L_t = P_t + N_t$$

be the Zariski decomposition on $\mathrm{Bl}_x(E \times E)$. Assume that $t \geq 0$ and that L_t is pseudo-effective. Then

$$E_x \not\subseteq \mathrm{Supp}(N_t).$$

In particular, this applies for every $\varepsilon_x < t < \mu_x$.

How it is used. Below ε_x , Claim 1.4 gives $N_t = 0$. Above ε_x , the negative part may be nonzero, but Claim 1.5 says that it never contains the exceptional divisor itself, as long as L_t is pseudo-effective.

1.4 Further proof-related claims

The following statements are also kept as claims, because Section 2 contains their proofs or proof-related remarks.

Claim 1.6 (the Kollár–Lazarsfeld divisor has Seshadri constant one). *For every point $x \in S$,*

$$\varepsilon(A_n; x) = 1.$$

Consequently, if $x \in S \setminus B$, then for every infinitesimal flag over x ,

$$\varepsilon'_x = \varepsilon_x = 1.$$

Claim 1.7 (the second eigensheaf summand vanishes in degree n). *The pullback map*

$$H^0(S, \mathcal{O}_S(nA_n)) \longrightarrow H^0(X, \mathcal{O}_X(nD_n))$$

is an isomorphism. Hence nD_n is not very ample.

Claim 1.8 (the complete linear series upstairs cannot equal the downstairs polygon).
The complete-series equality

$$\Delta_{(E_1, z_1)}(\pi_1^* D_n) = \Delta_{(E_x, z)}(\pi_x^* A_n)$$

would force $A_n^2 = 0$. Hence, for ample A_n , this equality is impossible.

Claim 1.9 (branch points require a weighted local valuation). Assume $x \in B$, B is smooth at x , and y is the unique point of X over x . Choose formal parameters (u, v) at x with $B = (v = 0)$. Then, after completing at x and y ,

$$\widehat{\mathcal{O}}_{X, y} \simeq k[[u, w]], \quad v = w^2.$$

Thus, for

$$f(u, v) = \sum_{i, j} a_{ij} u^i v^j,$$

one has

$$\pi^* f(u, w) = \sum_{i, j} a_{ij} u^i w^{2j}, \quad \text{ord}_y(\pi^* f) = \min\{i + 2j \mid a_{ij} \neq 0\}.$$

Equivalently, the ramified local model gives the $(1, 2)$ -weighted valuation in (u, v) , not the ordinary infinitesimal valuation on S .

Claim 1.10 (branch-point positivity follows from the surface theorem on X). Let $y \in X$ lie over a smooth branch point $x \in B$, and assume X is a smooth projective surface. For any infinitesimal flag over y , the largest inverted simplex constant of $D_n = \pi^* A_n$ at y equals

$$\varepsilon(D_n; y) = \varepsilon(\pi^* A_n; y).$$

Moreover, if

$$\pi_y : \text{Bl}_y X \rightarrow X, \quad E_y := \pi_y^{-1}(y), \quad L_t^X := \pi_y^* D_n - t E_y,$$

then the negative part of the Zariski decomposition of L_t^X vanishes for every

$$0 \leq t < \varepsilon(D_n; y).$$

1.5 Numerical quantities appearing in the draft

The manuscript uses the three numerical quantities

$$\mu_x = \sup\{t \mid \pi_x^* A_n - t E_x \text{ is big}\}, \quad \varepsilon_x = \sup\{t \mid \pi_x^* A_n - t E_x \text{ is ample}\},$$

and

$$\varepsilon'_x = \sup\left\{\xi \geq 0 \mid \Delta_\xi^{-1} \subseteq \Delta_{(E_x, z)}(\pi_x^* A_n)\right\}.$$

It also writes the Zariski decomposition

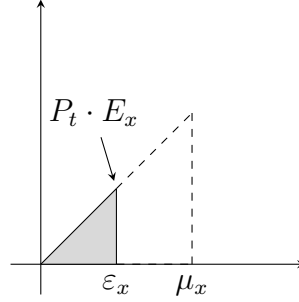
$$\pi_x^* A_n - t E_x = P_t + N_t.$$

Since $\pi_x^* A_n \cdot E_x = 0$ and $E_x^2 = -1$, one has

$$(\pi_x^* A_n - t E_x) \cdot E_x = t = P_t \cdot E_x + N_t \cdot E_x.$$

Thus the logical picture suggested by the draft is the following:

1. prove the upper-right boxed geometric claim, namely Claim 1.1;
2. justify the étale comparison only in the precise form stated in Claim 1.3;
3. control the lower part of the polygon by showing $N_t = 0$ for $t < \varepsilon_x$, as in Claim 1.4;
4. record that for pseudo-effective L_t with $t \geq 0$, the exceptional divisor E_x itself is never a component of N_t , as in Claim 1.5;
5. conclude the main positivity statement $\varepsilon'_x = \varepsilon_x$, namely Claim 1.2;
6. keep the branch-locus question “what if $x \in B$?” as a separate case, exactly as the draft suggests.



$$0 \leq t < \varepsilon_x \implies N_t = 0$$

2 Proofs of the claims and proof-related remarks

2.1 Proof of the second blow-up claim

Set

$$X_1 := \text{Bl}_{y_1}(X), \quad X_{12} := \text{Bl}_{y_1, y_2}(X), \quad \rho: X_{12} \rightarrow X_1.$$

Then

$$\pi_{1,2} = \pi_1 \circ \rho.$$

Fix a point $z_1 \in E_1$. Since the second blow-up is an isomorphism in a neighborhood of E_1 , the point z_1 determines a corresponding point

$$\tilde{z}_1 \in \tilde{E}_{y_1}.$$

Proposition 2.1 (the second blow-up is invisible to the infinitesimal flag over y_1). *With the notation above, the infinitesimal Newton–Okounkov polygons agree:*

$$\Delta_{(\tilde{E}_{y_1}, \tilde{z}_1)}(\pi_{1,2}^* D_n) = \Delta_{(E_1, z_1)}(\pi_1^* D_n).$$

Proof. Let

$$X_1 = \text{Bl}_{y_1}(X), \quad X_{12} = \text{Bl}_{y_1, y_2}(X), \quad \rho: X_{12} \rightarrow X_1.$$

Then $\pi_{1,2} = \pi_1 \circ \rho$. The point blown up by ρ is the point of X_1 lying over y_2 , while the flag

$$X_1 \supset E_1 \supset \{z_1\}$$

lies over y_1 . Since $y_1 \neq y_2$, we may choose an open neighborhood $U \subset X_1$ of E_1 avoiding the center of the second blow-up. A blow-up is an isomorphism away from its center, hence

$$\rho^{-1}(U) \simeq U.$$

Under this isomorphism $\tilde{E}_{y_1}$ is identified with E_1 , and $\tilde{z}_1$ with z_1 . This is the only geometric input needed from the second blow-up; see Hartshorne [11, Chapter II, §7] or the Stacks Project formulation [12].

For every $m \geq 0$, put $L_m = \mathcal{O}_{X_1}(m\pi_1^*D_n)$. Since $\pi_{1,2} = \pi_1 \circ \rho$, we have

$$\mathcal{O}_{X_{12}}(m\pi_{1,2}^*D_n) = \rho^*L_m.$$

Moreover $\rho_*\mathcal{O}_{X_{12}} = \mathcal{O}_{X_1}$, so pullback identifies $H^0(X_1, L_m)$ with $H^0(X_{12}, \rho^*L_m)$. Near the flag this identification is literally the identification of local equations through $\rho^{-1}(U) \simeq U$. Therefore a section s and its pullback ρ^*s have the same vanishing order along the first member of the flag, and after subtracting that first order they also have the same vanishing order at the marked point. Thus

$$\nu_{(\tilde{E}_{y_1}, \tilde{z}_1)}(\rho^*s) = \nu_{(E_1, z_1)}(s)$$

for every nonzero section in every degree.

The valuation semigroups are consequently identical, and the Newton–Okounkov polygons, being the closed convex hulls of the normalized valuation semigroups, are identical:

$$\Delta_{(\tilde{E}_{y_1}, \tilde{z}_1)}(\pi_{1,2}^*D_n) = \Delta_{(E_1, z_1)}(\pi_1^*D_n).$$

This uses only the standard functoriality of admissible-flag valuations under an isomorphism near the flag; compare [2, §1] and [4, §2]. $\square$

2.2 Proof of the étale-local comparison claim

Definition 2.2 (pullback graded linear series). *For each $m \geq 0$, set*

$$W_m := \pi^*H^0(E \times E, \mathcal{O}_{E \times E}(mA_n)) \subseteq H^0(X, \mathcal{O}_X(mD_n)), \quad W_\bullet := \bigoplus_{m \geq 0} W_m.$$

We call $W_\bullet$ the pullback graded linear series associated to A_n and π ; this notation is local to this note.

Choose corresponding infinitesimal flags

$$(E \times E)_{x^\bullet}: \text{Bl}_x(E \times E) \supset E_x \supset \{z\}, \quad X_{y_1^\bullet}: \text{Bl}_{y_1}(X) \supset E_1 \supset \{z_1\}.$$

Here $z_1 \in E_1$ is chosen to correspond to $z \in E_x$ under the formal-local identification induced by π at the selected point y_1 .

Proposition 2.3 (étale-local comparison and inclusion of graded series). *For every $m \geq 0$ and every nonzero section*

$$s \in H^0(E \times E, \mathcal{O}_{E \times E}(mA_n)),$$

one has

$$\nu_{(E_1, z_1)}(\pi^*s) = \nu_{(E_x, z)}(s).$$

Consequently,

$$\Delta_{(E_x, z)}(\pi_x^*A_n) = \Delta_{(E_1, z_1)}(W_\bullet) \subseteq \Delta_{(E_1, z_1)}(\pi_1^*D_n).$$

Proof. Because $x \notin B$, the morphism π is étale at the selected point y_1 . After fixing the residue-field identification over $\mathbb{C}$, the completed local rings are identified:

$$\widehat{\mathcal{O}}_{E \times E, x} \simeq \widehat{\mathcal{O}}_{X, y_1}.$$

Blowing up is local for this comparison, so the corresponding completed local blow-ups and their exceptional divisors are identified as well. We choose $z_1 \in E_1$ to be the point corresponding to $z \in E_x$. This is the standard formal-local consequence of étaleness; see [14] and Hartshorne [11, Chapter II, §7].

Let $s \in H^0(E \times E, \mathcal{O}_{E \times E}(mA_n))$ be nonzero. After trivializing $\mathcal{O}(mA_n)$ near x , the section s is represented by an element of $\widehat{\mathcal{O}}_{E \times E, x}$. Its pullback π^*s is represented by the corresponding element of $\widehat{\mathcal{O}}_{X, y_1}$. Hence the first infinitesimal valuation component, namely the order at the blown-up point, is the same. After removing this common first order, the induced initial forms on the identified exceptional divisors are also the same, so the second component at z and z_1 is the same. Therefore

$$\nu_{(E_1, z_1)}(\pi^*s) = \nu_{(E_x, z)}(s).$$

Thus the complete linear series of mA_n on S and the pullback graded series W_m on X have the same valuation vectors in every degree. This gives

$$\Delta_{(E_x, z)}(\pi_x^*A_n) = \Delta_{(E_1, z_1)}(W_\bullet).$$

Finally, $W_m \subseteq H^0(X, \mathcal{O}_X(mD_n))$ for every m , and inclusion of graded linear series gives inclusion of valuation semigroups and hence of Newton–Okounkov bodies:

$$\Delta_{(E_x, z)}(\pi_x^*A_n) = \Delta_{(E_1, z_1)}(W_\bullet) \subseteq \Delta_{(E_1, z_1)}(\pi_1^*D_n).$$

The last step is only an inclusion, not an equality with the complete series on X ; compare the definitions in [2, §1] and [4, §2]. $\square$

Remark 2.4. *The section-space inclusion is used only in the final step. The equality before it comes from étale-local identification of valuation vectors for the pullback graded linear series $W_\bullet$. Thus the argument compares the complete linear series on S with $W_\bullet$ on X , and it does not assert equality with the complete linear series on X .*

2.3 Proof of the largest inverted simplex claim

Definition 2.5 (two numbers used in the manuscript). *Fix $z \in E_x$ and set*

$$\varepsilon'_x := \sup \left\{ \xi \geq 0 \mid \Delta_\xi^{-1} \subseteq \Delta_{(E_x, z)}(\pi_x^*A_n) \right\}, \quad \varepsilon_x := \sup \{ t \mid \pi_x^*A_n - tE_x \text{ is ample} \}.$$

Here

$$\Delta_\xi^{-1} := \{ (t, y) \in \mathbb{R}_{\geq 0}^2 \mid 0 \leq t \leq \xi, 0 \leq y \leq t \}$$

is the two-dimensional inverted simplex of size ξ . The number ε'_x is the largest inverted simplex constant in this setting, while ε_x is the ample threshold used in the manuscript; only the symbols are specific to this note.

Proposition 2.6 (largest inverted simplex constant and Seshadri constant). *The value ε'_x is independent of z , and*

$$\varepsilon'_x = \varepsilon_x = \varepsilon(A_n; x).$$

Proof. By definition, ε'_x is precisely the largest inverted simplex constant of the infinitesimal Newton–Okounkov polygon $\Delta_{(E_x, z)}(\pi_x^* A_n)$. Since A_n is ample, in particular big, Küronya–Lozovanu’s surface theorem gives

$$\varepsilon'_x = \varepsilon(\|A_n\|; x).$$

For a big and nef divisor the moving Seshadri constant agrees with the usual Seshadri constant, so

$$\varepsilon'_x = \varepsilon(A_n; x).$$

These are exactly [4, Theorem 3.10 and Proposition 3.13].

It remains only to compare this usual Seshadri constant with the manuscript’s ample threshold

$$\varepsilon_x = \sup\{t \mid \pi_x^* A_n - tE_x \text{ is ample}\}.$$

The usual Seshadri constant is the nef threshold of $\pi_x^* A_n - tE_x$. Thus every ample value is automatically bounded by $\varepsilon(A_n; x)$. Conversely, take $0 < t < \varepsilon(A_n; x)$ and set

$$L_t = \pi_x^* A_n - tE_x.$$

For the exceptional curve E_x , one has $L_t \cdot E_x = t > 0$. For the strict transform of a curve not passing through x , the intersection is $A_n \cdot C > 0$. For the strict transform of a curve $C \ni x$, the curve formula for the Seshadri constant gives

$$A_n \cdot C - t \operatorname{mult}_x(C) > 0.$$

Hence L_t has positive intersection with every irreducible curve. Also

$$L_t^2 = A_n^2 - t^2 > 0,$$

because $t < \varepsilon(A_n; x) \leq \sqrt{A_n^2}$. Nakai–Moishezon therefore makes L_t ample. This proves that the ample and nef thresholds have the same supremum:

$$\varepsilon_x = \varepsilon(A_n; x).$$

Combining the equalities gives

$$\varepsilon'_x = \varepsilon(A_n; x) = \varepsilon_x.$$

The right-hand side does not depend on z , so neither does ε'_x . For the standard Seshadri and Nakai–Moishezon inputs, see Lazarsfeld [1, Chapters 1 and 5]. $\square$

2.4 Proof of the negative-part vanishing claim

For $t \geq 0$, write

$$L_t := \pi_x^* A_n - tE_x, \quad L_t = P_t + N_t$$

for the Zariski decomposition on $\operatorname{Bl}_x(E \times E)$.

Proposition 2.7 (the negative part vanishes below the Seshadri constant). *If $0 \leq t < \varepsilon(A_n; x)$, then*

$$N_t = 0.$$

Proof. For $0 \leq t < \varepsilon(A_n; x)$, the divisor

$$L_t = \pi_x^* A_n - tE_x$$

is nef by the nef-threshold definition of the Seshadri constant. Write its Zariski decomposition as

$$L_t = P_t + N_t, \quad N_t = \sum_{i=1}^r a_i C_i, \quad a_i > 0.$$

Assume, for contradiction, that $N_t \neq 0$. Orthogonality in the Zariski decomposition gives $P_t \cdot C_i = 0$ for each component C_i . Since $L_t = P_t + N_t$, it follows that

$$N_t \cdot C_i = L_t \cdot C_i \geq 0,$$

because L_t is nef. Multiplying by a_i and summing gives

$$N_t^2 = \sum_i a_i (N_t \cdot C_i) \geq 0.$$

But the intersection matrix of the components of the negative part is negative definite, so a nonzero N_t must satisfy $N_t^2 < 0$. This contradiction proves $N_t = 0$. The argument is the standard nef-versus-negative-definite argument in the surface Zariski decomposition; see Lazarsfeld [1, Chapter 2]. $\square$

Corollary 2.8. *For every $0 \leq t < \varepsilon_x$ one has*

$$L_t = P_t, \quad N_t = 0, \quad P_t \cdot E_x = t.$$

Proof. By Proposition 2.6, the manuscript's threshold satisfies

$$\varepsilon_x = \varepsilon(A_n; x).$$

Hence Proposition 2.7 applies for every $0 \leq t < \varepsilon_x$ and gives $N_t = 0$. The Zariski decomposition

$$L_t = P_t + N_t$$

therefore reduces to $L_t = P_t$. Intersecting with E_x gives

$$P_t \cdot E_x = L_t \cdot E_x = (\pi_x^* A_n - tE_x) \cdot E_x = t,$$

since $\pi_x^* A_n \cdot E_x = 0$ and $E_x^2 = -1$. These are the standard intersection formulas on the blow-up of a smooth surface at one point; see Hartshorne [11, Chapter V, §3]. $\square$

2.5 Proof of the exceptional-divisor exclusion claim

Proposition 2.9 (the exceptional divisor is not in the negative part). *Let*

$$L_t := \pi_x^* A_n - tE_x, \quad L_t = P_t + N_t$$

be the Zariski decomposition on $Y := \text{Bl}_x(E \times E)$. If $t \geq 0$ and L_t is pseudo-effective, then

$$E_x \not\subseteq \text{Supp}(N_t).$$

Consequently, for every $\varepsilon_x < t < \mu_x$, the exceptional divisor E_x is not an irreducible component of N_t .

Proof. Assume, for contradiction, that E_x occurs in N_t . Write

$$N_t = aE_x + N'_t, \quad a > 0,$$

with N'_t effective and without an E_x -component. Since components of the negative part are orthogonal to the positive part, $P_t \cdot E_x = 0$. If $N'_t = 0$, then

$$t = L_t \cdot E_x = (P_t + aE_x) \cdot E_x = -a < 0,$$

using $\pi_x^* A_n \cdot E_x = 0$ and $E_x^2 = -1$. This contradicts $t \geq 0$, so $N'_t \neq 0$.

Let $\Gamma_1, \dots, \Gamma_r$ be the irreducible components of N'_t , and set

$$C_i := \pi_x(\Gamma_i), \quad m_i := \text{mult}_x(C_i) = E_x \cdot \Gamma_i.$$

The intersection matrix of $E_x, \Gamma_1, \dots, \Gamma_r$ is a principal submatrix of the negative-definite intersection matrix of $\text{Supp}(N_t)$, hence is negative definite. If $M = (C_i \cdot C_j)$ and $m = (m_1, \dots, m_r)^T$, then the blow-up intersection formulas give

$$Q = \begin{pmatrix} -1 & m^T \\ m & M - mm^T \end{pmatrix}.$$

Since Q is negative definite, its Schur complement along the upper-left block (-1) is also negative definite. That Schur complement is exactly M .

This is impossible on the abelian surface $E \times E$. Indeed, every irreducible effective curve on an abelian surface is nef, so each $C_i^2 \geq 0$. A negative-definite matrix must have strictly negative diagonal entries, whereas the diagonal entries of M are precisely the numbers C_i^2 . The contradiction proves that E_x cannot occur in N_t . The final assertion follows because $\varepsilon_x < t < \mu_x$ lies in the big, hence pseudo-effective, range. See Lazarsfeld [1, Chapters 1 and 2] for the nefness of effective curves on abelian surfaces and the surface Zariski-decomposition facts. $\square$

2.6 Proof of the Kollár–Lazarsfeld Seshadri claim

A_n denotes the Kollár–Lazarsfeld divisor on $S = E \times E$ for an arbitrary elliptic curve E and integer $n \geq 2$.

Proposition 2.10 (the Kollár–Lazarsfeld divisor has Seshadri constant one). *For every point $x \in S$,*

$$\varepsilon(A_n; x) = 1.$$

Consequently, if $x \in S \setminus B$, then for every infinitesimal flag over x ,

$$\varepsilon'_x = \varepsilon_x = 1.$$

Proof. Through every point $x \in E \times E$ there is a translate $F_{2,x}$ of F_2 . It is a smooth elliptic curve through x with multiplicity one. Since

$$A_n \cdot F_2 = n - (n - 1) = 1,$$

the curve formula for Seshadri constants gives

$$\varepsilon(A_n; x) \leq \frac{A_n \cdot F_{2,x}}{\text{mult}_x(F_{2,x})} = 1.$$

For the reverse inequality, use the standard lower bound on abelian varieties: an ample integral line bundle on an abelian variety has Seshadri constant at least 1, and by translation invariance the value is independent of the point. Thus $\varepsilon(A_n; x) \geq 1$. Therefore

$$\varepsilon(A_n; x) = 1$$

for every $x \in S$. If $x \notin B$, Proposition 2.6 identifies the two manuscript constants with this Seshadri constant, so

$$\varepsilon'_x = \varepsilon_x = 1.$$

See Lazarsfeld [1, Example 5.3.10]. □

2.7 Proof of the eigensheaf-summand claim

Here $S = E \times E$, $R = F_1 + F_2$, and $\pi : X \rightarrow S$ is the standard double cover branched along a smooth divisor $B \in |2R|$. We keep $D_n = \pi^*A_n$.

Proposition 2.11 (the second eigensheaf summand vanishes in degree n). *The pullback map*

$$H^0(S, \mathcal{O}_S(nA_n)) \longrightarrow H^0(X, \mathcal{O}_X(nD_n))$$

is an isomorphism. Hence nD_n is not very ample.

Proof. The double cover is defined by the square root $\mathcal{O}_S(R)$, so

$$\pi_*\mathcal{O}_X \simeq \mathcal{O}_S \oplus \mathcal{O}_S(-R).$$

By the projection formula,

$$H^0(X, \mathcal{O}_X(nD_n)) \simeq H^0(S, \mathcal{O}_S(nA_n)) \oplus H^0(S, \mathcal{O}_S(nA_n - R)).$$

Thus the pullback map is an isomorphism exactly when the second summand vanishes.

For the Kollár–Lazarsfeld divisor, the intersection computation gives

$$(nA_n - R)^2 = -2(n-1)^3 < 0.$$

On $E \times E$, every effective curve is nef, so a nonzero effective divisor has nonnegative self-intersection. Hence $nA_n - R$ cannot be effective, and therefore

$$H^0(S, \mathcal{O}_S(nA_n - R)) = 0.$$

This proves that every section of nD_n is pulled back from S .

Now take $x \in S \setminus B$ with $\pi^{-1}(x) = \{y_1, y_2\}$. A pulled-back section has the same value at y_1 and y_2 , so nD_n cannot separate these two points. Hence nD_n is not very ample. This is the obstruction used in Lazarsfeld [1, Example 1.5.7]; for the eigensheaf decomposition see also Pardini [10, §2]. □

2.8 Proof of the complete-series area claim

Assume A_n is ample and $D_n = \pi^*A_n$. Fix corresponding infinitesimal flags over a non-branch point $x \notin B$:

$$Y_x^\bullet : \text{Bl}_x(E \times E) \supset E_x \supset \{z\}, \quad X_{y_1}^\bullet : \text{Bl}_{y_1}(X) \supset E_1 \supset \{z_1\}.$$

Proposition 2.12 (the complete linear series on X gives twice the area). *The complete-series equality*

$$\Delta_{(E_1, z_1)}(\pi_1^* D_n) = \Delta_{(E_x, z)}(\pi_x^* A_n)$$

would force $A_n^2 = 0$. Hence, for ample A_n , the equality is impossible.

Proof. The area of a Newton–Okounkov polygon on a smooth surface is half the volume of the divisor. Since volume is unchanged by pulling back to the blow-up,

$$\text{Area}(\Delta_{(E_x, z)}(\pi_x^* A_n)) = \frac{\text{vol}(A_n)}{2} = \frac{A_n^2}{2}.$$

For the complete linear series upstairs, the finite map π has degree 2, so

$$D_n^2 = (\pi^* A_n)^2 = 2A_n^2.$$

Again using the surface volume formula for Newton–Okounkov polygons,

$$\text{Area}(\Delta_{(E_1, z_1)}(\pi_1^* D_n)) = \frac{D_n^2}{2} = A_n^2.$$

If the complete-series polygons on X and S were equal, their areas would be equal. This would force

$$\frac{A_n^2}{2} = A_n^2,$$

hence $A_n^2 = 0$. But A_n is ample on a surface, so $A_n^2 > 0$. Therefore the complete-series equality is impossible. The comparison in Proposition 2.3 is consequently the sharp one: equality holds only for the pullback graded series $W_\bullet$, not for the complete linear series on X . See [2, Theorem A] for the volume formula and Lazarsfeld [1, Chapter 1] for ampleness on surfaces. $\square$

Remark 2.13 (Section decomposition on the double cover). *The volume discrepancy has a concrete section-theoretic explanation.*

- (1) **Double-cover decomposition.** *In the present Kollár–Lazarsfeld setup the double cover is the standard cover defined by the square root $\mathcal{O}_{E \times E}(R)$, where $R = F_1 + F_2$ and $B \in |2R|$. Hence*

$$\pi_* \mathcal{O}_X \simeq \mathcal{O}_{E \times E} \oplus \mathcal{O}_{E \times E}(-R).$$

More generally, a double cover defined by a line bundle M has $\pi_ \mathcal{O}_X \simeq \mathcal{O}_{E \times E} \oplus M^{-1}$ and $M^{\otimes 2} \simeq \mathcal{O}_{E \times E}(B)$. This is the standard $\mathbb{Z}/2$ -cover construction and eigensheaf decomposition; see Pardini [10, §2]; the concrete $M = \mathcal{O}(R)$ case is exactly the one used in Lazarsfeld [1, Example 1.5.7].*

- (2) **Section decomposition.** *Projection formula gives*

$$H^0(X, \mathcal{O}_X(mD_n)) \simeq H^0(E \times E, \mathcal{O}(mA_n)) \oplus H^0(E \times E, \mathcal{O}(mA_n - R)).$$

- (3) **Interpretation.** *The first summand is exactly W_m . The second summand is the (-1) -eigenspace for the deck involution. Since R is fixed and A_n is ample, asymptotic Riemann–Roch on the abelian surface gives*

$$h^0(E \times E, \mathcal{O}(mA_n - R)) = \frac{A_n^2}{2} m^2 + O(m),$$

so the second eigenspace summand has the same leading m^2 -order as the first summand. This explains the factor of 2 in the volume calculation above.

2.9 Proof of the branch-local weighted valuation claim

Assume $x \in B$, that B is smooth at x , and let y be the unique point of X over x . Choose formal parameters (u, v) at x with

$$B = (v = 0).$$

For a local equation on S , write

$$f(u, v) = \sum_{i,j} a_{ij} u^i v^j.$$

Proposition 2.14 (branch points require a weighted local valuation). *After completing at x and y , the double cover has local form*

$$\hat{\mathcal{O}}_{X,y} \simeq k[[u, w]], \quad v = w^2.$$

Thus

$$\pi^* f(u, w) = \sum_{i,j} a_{ij} u^i w^{2j}, \quad \text{ord}_y(\pi^* f) = \min\{i + 2j \mid a_{ij} \neq 0\}.$$

Equivalently, the ramified local model corresponds to the $(1, 2)$ -weighted valuation in (u, v) , not to the ordinary infinitesimal valuation on S .

Proof. Since B is smooth at x , choose formal parameters (u, v) with $B = (v = 0)$. A double cover ramified along B , in characteristic different from 2, has completed local form

$$\hat{\mathcal{O}}_{X,y} \simeq k[[u, w]], \quad v = w^2,$$

after absorbing a unit square. This is the usual local normal form for a smooth double cover; see Pardini [10, §2].

For

$$f(u, v) = \sum_{i,j} a_{ij} u^i v^j,$$

pullback simply substitutes $v = w^2$:

$$\pi^* f(u, w) = \sum_{i,j} a_{ij} u^i w^{2j}.$$

The order at y , measured in the ordinary parameters (u, w) , is therefore

$$\text{ord}_y(\pi^* f) = \min\{i + 2j \mid a_{ij} \neq 0\}.$$

By contrast, the ordinary order of f at x would be $\min\{i + j \mid a_{ij} \neq 0\}$. Thus the branch case is not governed by the ordinary infinitesimal valuation on S ; it is governed by the $(1, 2)$ -weighted valuation in the variables (u, v) . $\square$

2.10 Proof of the branch-point positivity claim

Corollary 2.15 (branch-point positivity follows from the surface theorem on X). *Let $y \in X$ lie over a smooth branch point $x \in B$. Assume that X is a smooth projective surface. For any infinitesimal flag over y , the largest inverted simplex constant of $D_n = \pi^* A_n$ at y equals the Seshadri constant*

$$\varepsilon(D_n; y) = \varepsilon(\pi^* A_n; y).$$

Moreover, if

$$\pi_y : \mathrm{Bl}_y(X) \rightarrow X, \quad E_y = \pi_y^{-1}(y), \quad L_t^X := \pi_y^* D_n - tE_y,$$

then the negative part of the Zariski decomposition of L_t^X vanishes for every

$$0 \leq t < \varepsilon(D_n; y).$$

Proof. Apply K ronya–Lozovanu’s surface theorem directly on the smooth projective surface X . Since $D_n = \pi^* A_n$ is ample, the largest inverted simplex constant at y equals the moving Seshadri constant, and the moving Seshadri constant equals the ordinary one:

$$\xi(D_n; y) = \varepsilon(\|D_n\|; y) = \varepsilon(D_n; y) = \varepsilon(\pi^* A_n; y).$$

Here the ramification affects only comparisons with valuations downstairs on S , not the intrinsic positivity theorem on X ; see [4, Theorem 3.10 and Proposition 3.13].

For the final assertion, let

$$L_t^X = \pi_y^* D_n - tE_y$$

with $0 \leq t < \varepsilon(D_n; y)$. By definition of the Seshadri constant, L_t^X is nef. A nef divisor has trivial negative part in its Zariski decomposition, so the negative part of L_t^X is zero for every such t . See Lazarsfeld [1, Chapter 2]. $\square$

Remark 2.16 (What is still impossible without more data). *The branch case is now reduced to two independent pieces.*

- (1) **Local piece.** Proposition 2.14 gives the precise weighted valuation replacing the  tale comparison.
- (2) **Positivity piece.** Corollary 2.15 gives the correct Seshadri-constant statement on X .
- (3) **Missing numerical piece.** Although the divisor A_n on S is now fixed, a closed numerical value for the ramification-point Seshadri constant on X $\varepsilon(\pi^* A_n; y)$ cannot be obtained from the present file alone, because it can depend on the curve geometry of the chosen smooth branch divisor $B \in |2(F_1 + F_2)|$ and of the resulting double cover.

The statement justified by the present hypotheses is therefore Corollary 2.15, not an unsupported equality with the number $\varepsilon(A_n; x)$ on S .

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